

Ecosystems: Summary

Answers

Use the listed words to complete the sentences.

capture	consumers	energy	first	food
fungi	glucose	interbreed	matter	plastics
quadrat	recaptured	species	trophic	water

1. If there are many organisms or the area is large, the**quadrat**..... method may be used to estimate the population of a specific organism.
2. The capture–recapture method is used to estimate populations of animals. Some animals are captured and tagged, then released back into the environment. At a later stage the researchers return to the field and**capture**..... more animals. By working out what proportion of the**recaptured**..... animals are tagged, they can estimate the total number of animals.
3. Organisms that belong to the same species can**interbreed**..... to produce fertile offspring. Animals may look quite similar yet belong to different**species**.....
4. The source of**energy**..... in all ecosystems (except those that use geothermal energy) is sunlight. The light from the sun is converted to plant matter as plants carry out photosynthesis. Animals eat the plant matter and they take in the energy stored in plants. Bacteria and**fungi**..... obtain their energy from the breakdown of other organisms.
5. In the process of photosynthesis carbon dioxide and water combine to form**glucose**..... and oxygen. In cellular respiration, sugars such as glucose react with oxygen to form carbon dioxide and**water**.....
6. The flow of energy through ecosystems can be represented in a**food**..... web. The arrows in the food web show the direction in which the energy is transferred.
7. In a food web the plants or producers make up the first**trophic**..... level (feeding level). The animals that eat plants make up the next trophic level. They are called herbivores or first order consumers.
8. A large mass of producers is needed to support a small number of top level**consumers**..... This can be shown using a biomass pyramid.
9. Whilst energy flows through ecosystems**matter**..... is recycled. The water, carbon, nitrogen and phosphorus cycles ensure that matter is recycled.
10. Many materials can be recycled or composted rather than ending up in landfill. Glass, aluminium and other metals, paper and many**plastics**..... can be made into new objects rather than be dumped at the tip.